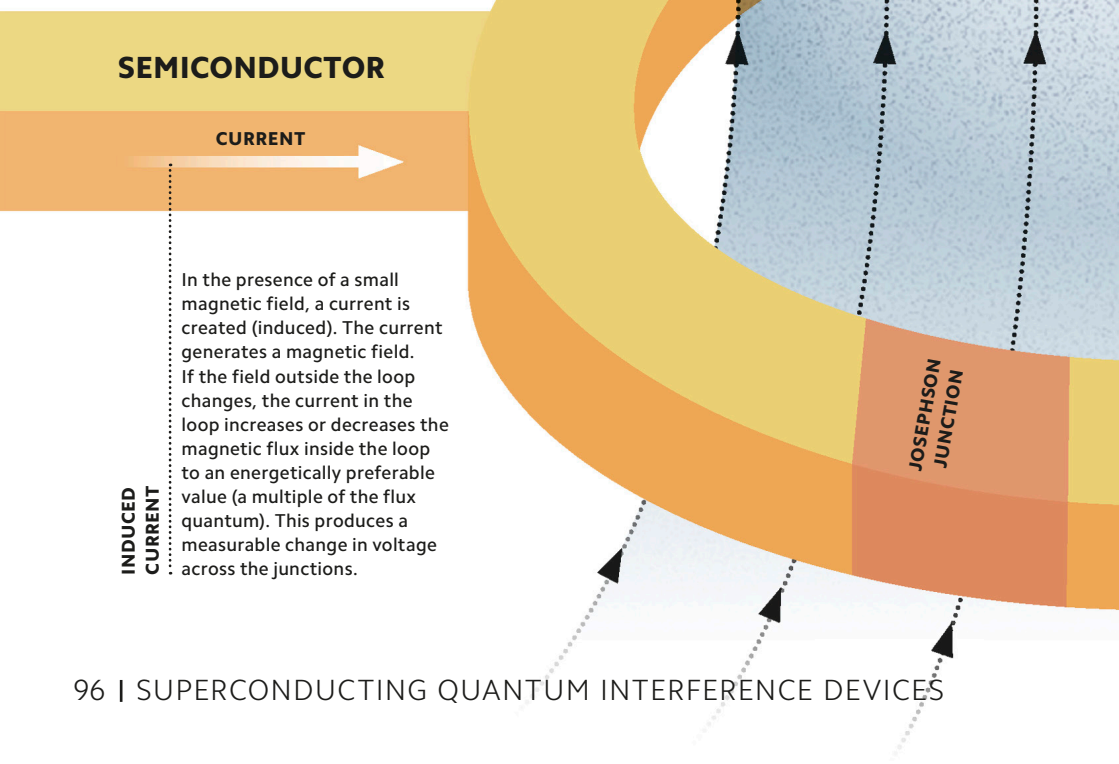


SQUIDS

A superconducting quantum interference device (SQUID) is used to detect extremely faint magnetic fields, including signals associated with neural activity. A SQUID is based on a superconducting loop containing two Josephson junctions (see p.95). In a magnetic field, the current in the ring varies to shift the magnetic flux (number of field lines) within it to an energetically preferable value, causing voltage to vary with the magnetic field.



In the presence of a small magnetic field, a current is created (induced). The current generates a magnetic field. If the field outside the loop changes, the current in the loop increases or decreases the magnetic flux inside the loop to an energetically preferable value (a multiple of the flux quantum). This produces a measurable change in voltage across the junctions.